

CHAPTER 13

Nuclei

Atomic Nucleus, Its Size, Density and Weight

1. A nucleus of mass number 189 splits into two nuclei having mass number 125 and 64. The ratio of radius of two daughter nuclei respectively is : (2022)
a. 25 : 16 b. 1 : 1 c. 4 : 5 d. 5 : 4
2. The energy equivalent of 0.5 g of a substance is : (2020)
a. 4.5×10^{13} J b. 1.5×10^{13} J
c. 0.5×10^{13} J d. 4.5×10^{16} J
3. If radius of the ${}^{27}_{12}\text{Al}$ nucleus is taken to be R_{Al} , then the radius of ${}^{125}_{53}\text{Te}$ nucleus is nearly: (2015)
a. $\frac{5}{3}R_{\text{Al}}$ b. $\frac{3}{5}R_{\text{Al}}$
c. $\left(\frac{13}{53}\right)^{1/3}R_{\text{Al}}$ d. $\left(\frac{53}{13}\right)^{1/3}R_{\text{Al}}$

Mass Energy, Nuclear Binding Energy and Nuclear Force

4. A nucleus with mass number 240 breaks into two fragments each of mass number 120, the binding energy per nucleon of unfragmented nuclei is 7.6 MeV while that of fragments is 8.5 MeV. The total **gain** in the Binding Energy in the process is: (2021)
a. 9.4 MeV b. 804 MeV
c. 216 MeV d. 0.9 MeV

Radioactive Decay Law, Half-Life and Average Life and Activity of a Radioactive Substance

5. Select the correct statements among the following :
A. Slow neutrons can cause fission in ${}^{235}_{92}\text{U}$ than fast neutrons.
B. α -rays are Helium nuclei.
C. β -rays are fast moving electrons or positrons.
D. γ -rays are electromagnetic radiations of wavelengths larger than X-rays.
Choose the most appropriate answer from the options given below: (2024 Re)
a. A, B and C only b. A, B and D only
c. A and B only d. C and D only

6. The half life of a radioactive substance is 20 minutes. In how much time, the activity of substance drops to $(1/16)^{\text{th}}$ of its initial value? (2023)
a. 80 minutes b. 20 minutes
c. 40 minutes d. 60 minutes

7. Given below are two statements: (2022 Re)

Statement I: The law of radioactive decay states that the number of nuclei undergoing the decay per unit time is inversely proportional to the total number of nuclei in the sample.

Statement II: The half life of a radionuclide is the sum of the life time of all nuclei, divided by the initial concentration of the nuclei at time $t = 0$.

In the light of the above statements, choose the most appropriate answer from the options given below:

- a. Statement I is incorrect but Statement II is correct.
 - b. Both Statement I and Statement II are correct.
 - c. Both Statement I and Statement II are incorrect.
 - d. Statement I is correct but Statement II is incorrect.
8. At any instant, two elements X_1 and X_2 have same number of radioactive atoms. If the decay constant of X_1 and X_2 are 10λ and λ respectively, then the time when the ratio of their atoms becomes $\frac{1}{e}$ respectively will be: (2022 Re)
a. $\frac{1}{5\lambda}$ b. $\frac{1}{11\lambda}$ c. $\frac{1}{9\lambda}$ d. $\frac{1}{8\lambda}$

9. The half life of a radioactive sample undergoing α -decay is 1.4×10^{17} s. If the number of nuclei in the sample is 2.0×10^{21} , the activity of the sample is nearly. (2020-Covid)
a. 10^5 Bq b. 10^6 Bq c. 10^3 Bq d. 10^4 Bq
10. For a radioactive material, half-life is 10 minutes. If initially there are 600 number of nuclei, the time taken (in minutes) for the disintegration of 450 nuclei is (2018)
a. 30 b. 10 c. 20 d. 15
11. Radioactive material 'A' has decay constant ' 8λ ' and material 'B' has decay constant ' λ '. Initially they have same number of nuclei. After what time, the ratio of number of nuclei of material 'A' to that of 'B' will be $1/e$? (2017-Delhi)
a. $\frac{1}{7\lambda}$ b. $\frac{1}{8\lambda}$ c. $\frac{1}{9\lambda}$ d. $\frac{1}{\lambda}$

12. The half-life of a radioactive substance is 30 minutes. The time (in minutes) taken between 40% decay and 85% decay of the same radioactive substance is: (2016 - II)
a. 45 b. 60 c. 15 d. 30
13. A radio isotope X with a half life of 1.4×10^9 years decays to Y which is stable. A sample of the rock from a cave was found to contain X and Y in the ratio 1 : 7. The age of the rock is: (2014)
a. 1.96×10^9 years b. 3.92×10^9 years
c. 4.20×10^9 years d. 8.40×10^9 years
14. The half life of a radioactive isotope ' X ' is 20 years. It decays to another element ' Y ' which is stable. The two elements ' X ' and ' Y ' were found to be in the ratio 1 : 7 in a sample of a given rock. The age of the rock is estimated to be: (2013)
a. 100 years b. 40 years
c. 60 years d. 80 years
16. A radioactive nucleus A_ZX undergoes spontaneous decay in the sequence
 ${}^A_ZX \rightarrow {}^{A-1}_{Z-1}B \rightarrow {}^{A-3}_{Z-3}C \rightarrow {}^{A-2}_{Z-2}D$, where Z is the atomic number of element X . The possible decay particles in the sequence are: (2021)
a. α, β^+, β^- b. β^+, α, β^- c. β^-, α, β^+ d. α, β^-, β^+
17. When a uranium isotope ${}^{235}_{92}\text{U}$ is bombarded with a neutron, it generates ${}^{89}_{36}\text{Kr}$, three neutrons and : (2020)
a. ${}^{91}_{40}\text{Zr}$ b. ${}^{101}_{36}\text{Kr}$ c. ${}^{103}_{36}\text{Kr}$ d. ${}^{144}_{56}\text{Br}$
18. The binding energy per nucleon of ${}^7_3\text{Li}$ and ${}^4_2\text{He}$ nuclei are 5.60 MeV and 7.06 MeV, respectively. In the nuclear reaction ${}^7_3\text{Li} + {}^1_1\text{H} \rightarrow {}^4_2\text{He} + {}^4_2\text{He} + Q$ the value of energy Q released is: (2014)
a. 19.6 MeV b. -2.4 MeV
c. 8.4 MeV d. 17.3 MeV

Nuclear Reactions

15. In the given nuclear reaction, the element X is : (2022)
 ${}^{22}_{11}\text{Na} \rightarrow X + e^+ + \nu$
a. ${}^{22}_{12}\text{Mg}$ b. ${}^{22}_{11}\text{Na}$ c. ${}^{22}_{10}\text{Ne}$ d. ${}^{22}_{10}\text{Ne}$

Nuclear Fission and Nuclear Fusion

19. A certain mass of Hydrogen is changed to Helium by the process of fusion. The mass defect in fusion reaction is 0.02866 u. The energy liberated per u is (given 1 u = 931 MeV): (2013)
a. 13.35 MeV b. 2.67 MeV
c. 26.7 MeV d. 6.675 MeV

Answer Key

1. (d) 2. (a) 3. (a) 4. (c) 5. (a) 6. (a) 7. (c) 8. (c) 9. (d) 10. (c)
11. (a) 12. (b) 13. (c) 14. (c) 15. (d) 16. (b) 17. (d) 18. (d) 19. (d)